

COURSE SYLLABUS

Linux Administration Essentials

COURSE NUMBER

LNK-001

DURATION

1 day · 4 modules

AUDIENCE LEVEL

Beginner

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This course teaches the fundamental concepts and practical skills required to navigate, manage, and secure Linux operating systems. It is designed for aspiring system administrators and IT professionals with little to no prior Linux experience. Afterward, learners will be able to confidently perform basic command-line operations, manage user accounts, and configure essential system services.

DELIVERY FORMAT OPTIONS

Standard Instructor-Led Training (ILT). The instructor teaches each module, leads explanations and discussions, and guides students through the standard module labs using the published module timing.

Bootcamp / Lab-Centered Delivery. When Bootcamp Mode capstone labs are included, they provide an alternative delivery configuration, not additional required ILT lecture time. In this configuration, students primarily work through module labs and capstone lab scenarios while the trainer acts as a facilitator, answering questions, clarifying concepts, and helping students work through lab challenges.

Training providers may choose either the normal ILT model or the Bootcamp/lab-centered model based on audience readiness, delivery goals, and available class time.

AT COURSE COMPLETION

After completing this course, you will be able to:

Navigate the Linux directory structure and perform basic file operations using the command-line interface.

Manage user accounts, groups, and file permissions to secure system resources.

Monitor system processes and manage essential services using standard system utilities.

Configure basic network settings and perform software package installations.

PREREQUISITES

- Basic familiarity with general computer concepts and operating system navigation
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COURSE OUTLINE

MODULE 1 Command-Line Navigation and File Management

90 min

Learn the essentials of navigating the Linux filesystem hierarchy and manipulating directories and files directly from the terminal.

LESSONS

Lesson 1: The Linux Filesystem and Directory Structure — 15 min

Lesson 2: Executing Fundamental Shell Commands — 15 min

Lesson 3: Module Review and Knowledge Assessment — 15 min

LAB EXERCISES

Navigating Files and Managing Directories — 35 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 90 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Identify structural components of the Linux filesystem hierarchy
- Execute essential terminal navigation commands to locate resources
- Create, duplicate, rename, and delete directories and files safely

MODULE 2 User Administration and Permissions Control

90 min

Gain absolute command over system security settings, authorization layers, user profile structures, and access controls.

LESSONS

Lesson 1: Managing User Accounts and Groups — 15 min

Lesson 2: Assigning File Permissions and Ownership — 15 min

Lesson 3: Module Review and Knowledge Assessment — 15 min

LAB EXERCISES

Configuring Users and Locking Down Resources — 35 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 90 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Provision and configure new user accounts and operational groups
- Analyze and set file permissions using numeric and symbolic notation
- Delegate elevated command execution authority utilizing sudo

MODULE 3 System Process and Service Management

90 min

Understand how to trace system background workloads, resolve resource contention, and control continuous running processes.

LESSONS

Lesson 1: Process Monitoring and Priority Control — 15 min

Lesson 2: Controlling System Daemons with systemd — 15 min

Lesson 3: Module Review and Knowledge Assessment — 15 min

LAB EXERCISES

Diagnosing Processes and Controlling Services — 35 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 90 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Track active system resources and run-time process statuses

Control processes using kill signals and process priority utilities

Manipulate services and system states through the systemd framework

MODULE 4 Network Configuration and Package Installation

90 min

Master essential networking diagnostics, interface management, and native software package lifecycles.

LESSONS

Lesson 1: Interpreting
Network Metrics
and Connectivity — 15 min

Lesson 2: Software
Distribution and
Package
Management — 15 min

Lesson 3: Module Review
and Knowledge
Assessment — 15 min

LAB EXERCISES

Enabling Network Interfaces
and Deploying Applications — 35 min

TIMING

Review / Summary: 10 min · 4 questions **Module total:** 90 min

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Inspect network interfaces and verify end-to-end IP connectivity
- Resolve software packages and resolve dependencies using repository managers
- Deploy and update standard system utility files safely

BOOTCAMP MODE — CAPSTONE LABS

Capstone labs are included for Bootcamp/lab-centered delivery. They are not additional required lecture time for Standard Instructor-Led Training. Use them when the client chooses a facilitated lab delivery model where the trainer coaches students through integrated practice scenarios.

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE Modules 1, 2, 3, 4

SKILLS TO BE INTEGRATED

- Navigate directories and perform basic command-line file operations

- Create user accounts, define groups, and enforce precise ownership and permission structures

- Inspect active processes and configure critical background system services

- Configure network settings and install system utilities using package managers

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Courseware Publisher download package.